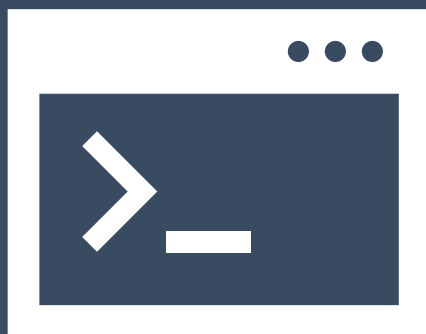


# Introduction to Spatial Transcriptomics

<https://tinyurl.com/hbc-spatial>



Harvard Chan Bioinformatics Core



# **Introductions!**





Shannan Ho Sui  
*Director*



James Billingsley



Emma Berdan



Elizabeth  
Partan



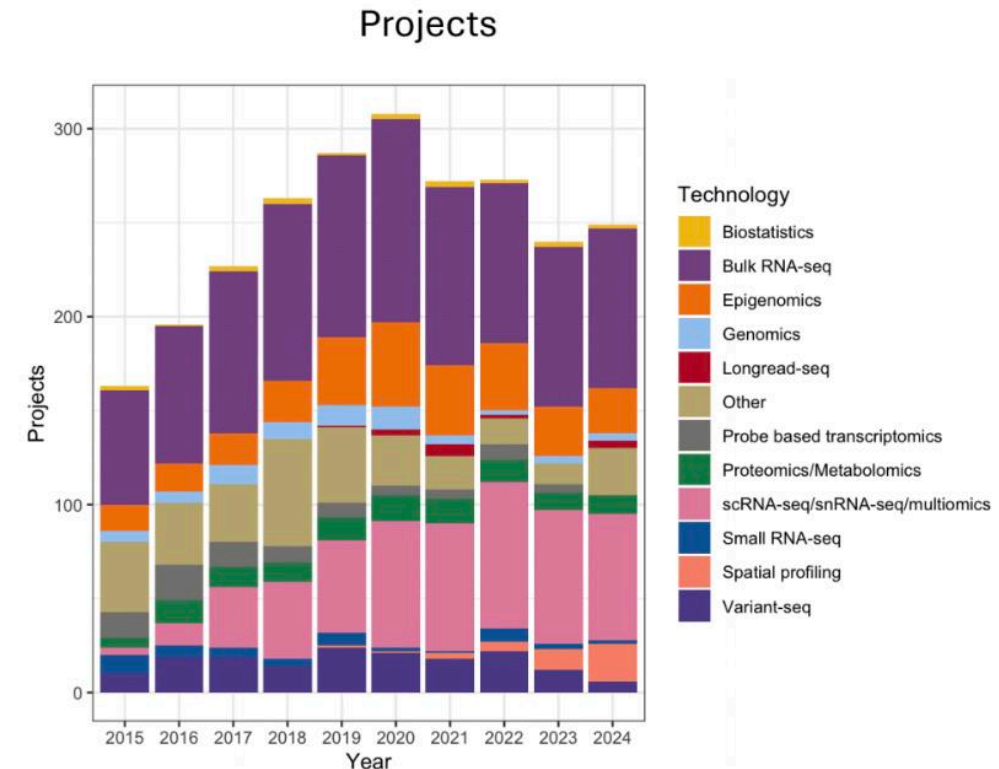
Will Gammerdinger



Noor Sohail

# Consulting

- Transcriptomics: Bulk, single cell, small RNA
- Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- Variant discovery: WGS, resequencing, exome-seq and CNV
- Multiomics integration
- Spatial biology
- Experimental design and grant support



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**HARVARD**  
**T.H. CHAN**  
SCHOOL OF PUBLIC HEALTH

NIEHS

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THE HARVARD CLINICAL  
AND TRANSLATIONAL  
SCIENCE CENTER

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**HARVARD**  
MEDICAL SCHOOL

# Training

- Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design
  - Basic Data Skills
    - Shell
    - R
    - Python
  - Advanced Topics: Analysis of high-throughput sequencing data
    - Chromatin Biology
    - Bulk RNA-seq
    - Differential Gene Expression
    - scRNA-seq
    - Variant Calling
    - Spatial Transcriptomics
    - Tools for Reproducible Research
  - Current Topics in Bioinformatics

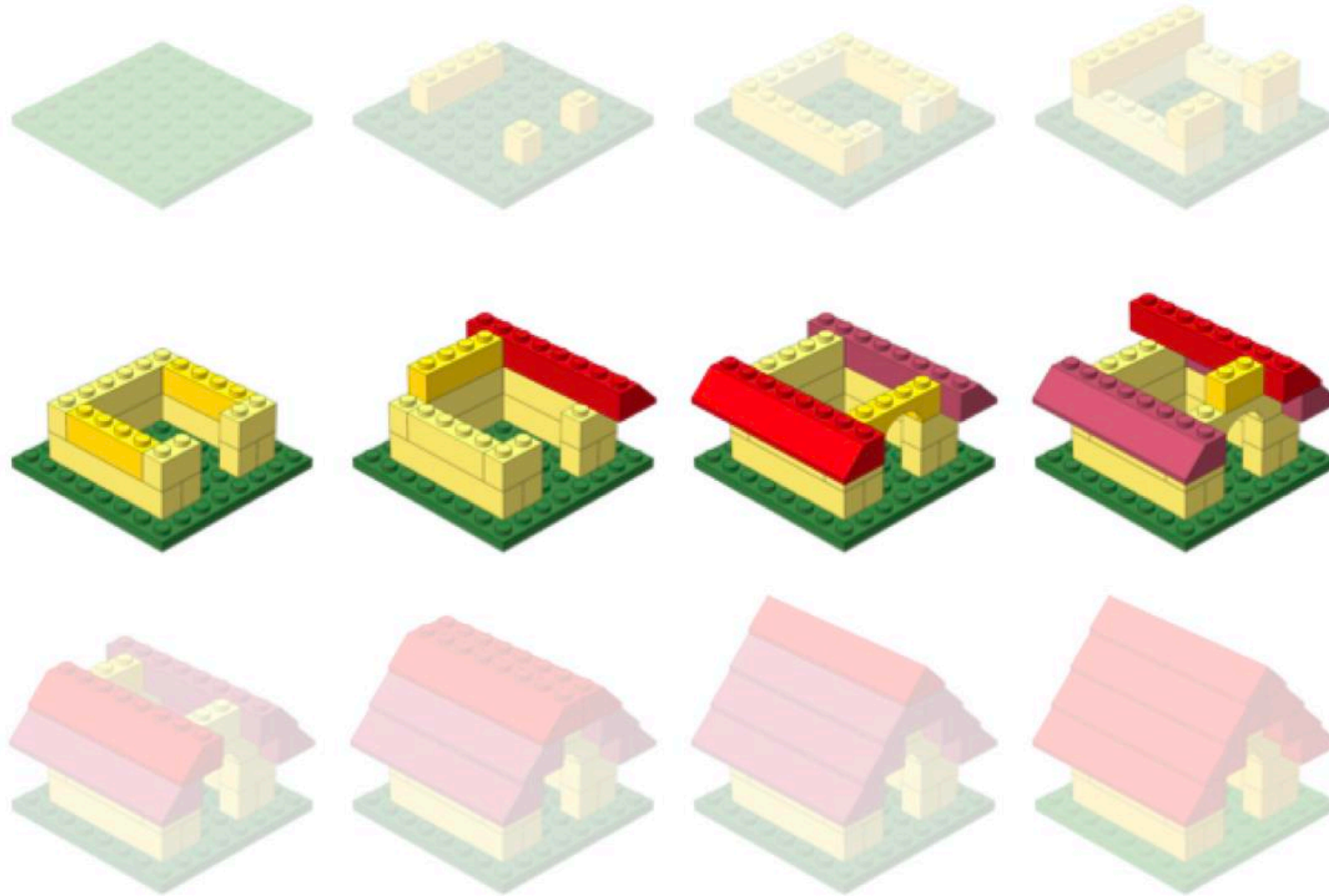
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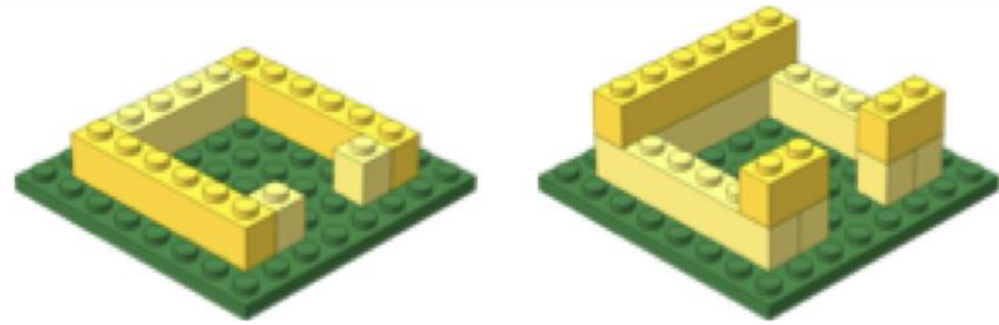
<https://hsph.harvard.edu/research/bioinformatics/training/>

# **Workshop scope**



Bioinformatic Data Analysis

# Learning Objectives



- Experimental design best practices of current spatial transcriptomics technologies
- Data loading and quality control of Visium HD samples
- Standard single-cell workflows adapted for spatial data, including highly variable gene selection, PCA, UMAP, kNN, and clustering
- Cell type annotation with deconvolution using RCTD for sequencing-based technologies
- Spatial-specific analyses that utilize the physical location of bins/cells on the tissue
  - Spatial clustering with BANKSY
  - Spatially variable gene detection with Moran's I
  - Cell-cell communication with CellChat

# Logistics



# Course schedule

Spatial Transcriptomics

Schedule

HBC

GitHub

Contact us

## Schedule - Spatial Transcriptomics

</> Code

### Pre-reading

1. [Spatial Technology Overview](#)
2. [Space Ranger Summary](#)
3. [Download Project](#)

### Day 1

| Time          | Topic                                                       | Instructor                          |
|---------------|-------------------------------------------------------------|-------------------------------------|
| 10:00 - 10:15 | <a href="#">Workshop Introduction</a>                       | Will                                |
| 10:15 - 11:15 | <a href="#">Introduction to Spatial Transcriptomics</a>     | <a href="#">Dr. Arpita Kulkarni</a> |
| 11:15 - 11:20 | Break                                                       |                                     |
| 11:20 - 11:55 | <a href="#">Load Visium HD Data</a>                         | Noor                                |
| 11:55 - 12:00 | Overview of self-learning materials and homework submission | Will                                |

<https://tinyurl.com/hbc-spatial>

# Course materials

- We continuously update our materials to reflect changes in the field/software

[Spatial Transcriptomics](#) [Schedule](#) [HBC](#) [GitHub](#) [Contact us](#) 

Pre-reading: 

[Spatial Transcriptomics Technologies](#)

Space Ranger Processing

Day 1: 

Day 1 Self-learning: 

Day 2: 

Day 2 Self-learning: 

Day 3: 

Day 3 Self-learning: 

Day 4: 

## Spatial Transcriptomics Technologies

[SPATIAL TRANSCRIPTOMICS](#) [EXPERIMENTAL DESIGN](#)

Explore the current landscape of spatial transcriptomics technologies. Focusing on contrasting sequencing-based and imaging-based platforms to show how they differ in resolution, throughput and sensitivity. By the end, you will be informed on which technology is best suited for your biological question.

KEYWORDS  
Imaging-based, Sequencing-based, Visium, MERFISH, seqFISH

Approximate time: 30 minutes

### Learning objectives

In this lesson, we will:

- Describe the evolution of high-throughput sequencing

On this page

[Learning objectives](#)

[Overview of lesson](#)

[High-throughput sequencing overview](#)

[Sequencing-based technologies](#)

[Imaging-based technologies](#)

[Summary of comparison](#)

[Experimental design considerations](#)

# Single Screen & 3 Windows

The image is a composite of three overlapping windows, illustrating a 'Single Screen & 3 Windows' setup. The top window is a Zoom meeting interface showing a video call with two participants, a chat window on the right, and a meeting controls bar at the bottom. The bottom-left window is a JupyterLab interface displaying a Python notebook with code for reading and processing CSV data. The bottom-right window is an RStudio interface showing a file explorer, a console with R version information, and a package manager.

**JupyterLab Window:**

```
[28]: metadata = pd.read_csv("data/mouse_exp_design.csv")

[29]: # Get the row names
metadata.index
# Get the column names
metadata.columns

[30]: Index(['sample1', 'sample2', 'sample3', 'sample4', 'sample5', 'sample6',
        'sample7', 'sample8', 'sample9', 'sample10', 'sample11', 'sample12'],
        dtype='object')

[31]: metadata.loc[0, :]

[32]: genotype    WT
        celltype   typek
        replicate      3
        Name: sample1, dtype: object

[33]: metadata.loc[:, 0]
```

**RStudio Window:**

R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"  
Copyright (C) 2026 The R Foundation for Statistical Computing  
Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

**RStudio File Explorer:**

| Name                          | Size   | Modified              |
|-------------------------------|--------|-----------------------|
| ..                            |        |                       |
| data                          |        |                       |
| figures                       |        |                       |
| README.txt                    | 1.9 KB | Apr 29, 2026, 8:39 PM |
| results                       |        |                       |
| scripts                       |        |                       |
| spatial_transcriptomics.Rproj | 205 B  | Apr 30, 2026, 1:12 PM |

*Our  
Recommendation*

# Single Screen & 3 Windows

**Zoom**

Meeting: Will Gammerding's screen

Participants: Noor Sohal, Will Gammerding (host)

**JupyterLab**

localhost:8881/lab/tree/Desktop/ntra\_python

File Explorer:

- data
- figures
- results
- data.zip
- ntra\_python.py

Code Editor:

```
[28]: metadata = pd.read_csv("data/mouse_exp_design.csv")

[29]: # Get the row names
      metadata.index
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[30]: Index(['sample1', 'sample2', 'sample3', 'sample4', 'sample5', 'sample6',
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[34]: sample1    WT
      sample2    WT
      sample3    WT
      sample4    KO
      sample5    KO
      sample6    KO
      sample7    WT
      sample8    WT
      sample9    KO
      sample10   KO
      sample11   KO
      sample12   KO
      Name: genotype, dtype: object
```

**Zoom Meeting**

You to Everyone 8:38 AM

I have a question about how to make figures in Python. How do you make multiple plots at once?

**bioinformatics**

On this page

- Learning objectives
- Overview of lesson
- High-throughput sequencing overview
- Sequencing-based technologies
- Imaging-based technologies
- Summary of comparison
- Experimental design considerations

**RStudio**

Files | Plots | Packages | Help | Viewer | Presentation

Folder | File | Delete | Rename | Settings

g > Training\_programs > Workshops > new\_workshop\_data > spatial\_transcriptomics

| Name                          | Size   | Modified              |
|-------------------------------|--------|-----------------------|
| ..                            |        |                       |
| data                          |        |                       |
| figures                       |        |                       |
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1:1 (Top Level) | Posit AI: No completions available. | R Script

Console | Terminal | Background Jobs

R - R 4.5.3 · ~/Harvard University Dropbox/HBC Team Folder (1)/Teaching/Training\_programs/Wc

R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"  
Copyright (C) 2026 The R Foundation for Statistical Computing  
Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
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Natural language support but running in an English locale

**Our Recommendation**

# Single Screen & 3 Windows

**Web Browser**

The image illustrates a single-screen workflow for a Zoom meeting. The top-left window displays a JupyterLab interface with a file explorer on the left, a code editor in the center, and a console at the bottom. The top-right window shows a web browser displaying a page titled 'transcriptomics' with a sidebar containing 'Learning objectives' and 'On this page'. The bottom window shows an R console output with R version 4.5.3 and a file explorer showing a directory structure.

**Our Recommendation**

# Single Screen & 3 Windows

**Our Recommendation**

# Single Screen & 3 Windows

The image displays three overlapping software windows, each with a distinct colored border and a large text label:

- Zoom** (Blue border): A Zoom meeting window titled "Will Gammerdinger's screen". It shows a video call with participants Noor Sohal and Will Gammerdinger. Below the video is a JupyterLab interface with a file explorer on the left and a code editor on the right. The code editor contains Python code for reading and processing a CSV file.
- Web Browser** (Green border): A web browser window showing a page titled "omics". The page has a red header with links for "HBC", "GitHub", and "Contact us". The main content area discusses transcriptomics technologies, focusing on contrasting single-cell RNA-seq and spatial transcriptomics. A sidebar on the right lists "Learning objectives" such as "Overview of lesson", "High-throughput sequencing overview", and "Sequencing-based technologies".
- R Studio** (Purple border): An R Studio window showing the R console and file explorer. The console displays the R version (4.5.3) and the path to the R project. The file explorer on the right shows a list of files and folders, including "figures", "README.txt", "results", "scripts", and "spatial\_transcriptomics.Rproj".

**Our Recommendation**

# Course participation

- ❑ Mandatory review of self-learning lessons and assignments
- ❑ Attendance required for all classes
- ❑ Your questions and active participation drive learning
- ❑ **We look forward to all your questions!**



# Course participation

- ☐ At-home lessons and exercises after each session
- ☐ Cover material not previously discussed
- ☐ Provides us feedback to help pace the course appropriately
- ☐ 3-5 hours to complete
- ☐ Homework load is heavier near the end of the workshop

# Using AI for Assignments

## ☐ Do

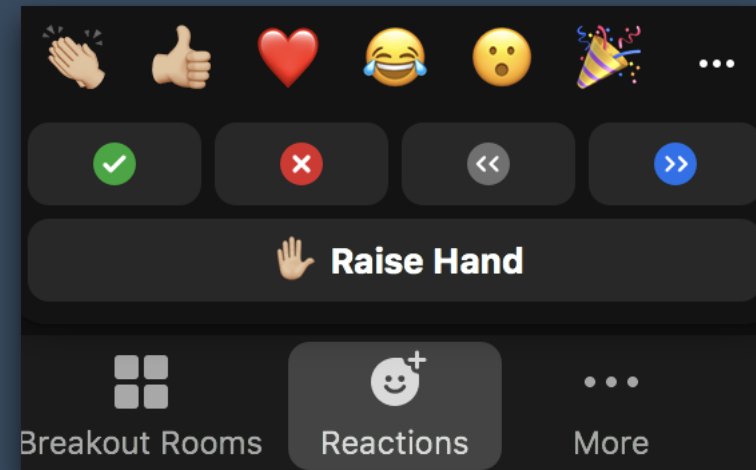
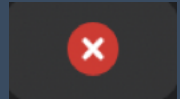
- ☐ Try to resolve error messages with it
- ☐ Test code written by AI on a dataset where you have expected results
- ☐ Take the time to review the generated code line-by-line

## ☐ Don't

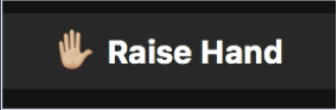
- ☐ Implement it in replacement to learning
- ☐ Write code that you don't understand
- ☐ Assume the output from an AI process is correct

# Odds & Ends

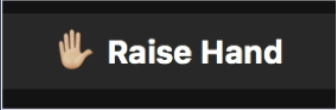
- ❑ Quit/minimize all applications that are not required for class
- ❑ Are you all set?
  - ❑ = "agree", "I'm all set"
  - ❑  = "disagree", "I need help"



# Odds & Ends

- Questions for the presenter?
  - Post the question in the Chat window OR
  -  when the presenter asks for questions
  - Let the Moderator know

# Odds & Ends

- Questions for the presenter?
  - Post the question in the Chat window OR
  -  when the presenter asks for questions
  - Let the Moderator know
- Technical difficulties with software?
  - Start a private chat with the Moderator with a description of the problem

# Contact Us

- *HBC training team:* [hbctraining@hsph.harvard.edu](mailto:hbctraining@hsph.harvard.edu)
- *HBC consulting:* [bioinformatics@hsph.harvard.edu](mailto:bioinformatics@hsph.harvard.edu)